Felix Zhang

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Education

University of Toronto Sept. 2025 – Jan. 2027
Master of Science in Computer Science *4.0/4.0 cGPA*
Advisor: Prof. Qizhen Zhang

University of Toronto Sept. 2021 – June 2025
Honours Bachelor of Science in Computer Science; Major in Mathematics *3.96/4.0 cGPA*

Publications

EFL: Efficient, Fault-tolerant, and Low-cost Shuffle Service with Complete DPU Offloading
Felix Zhang, Joseph Zhu, and Qizhen Zhang
Under Review

PD3: Prefetching Data with DPUs for Disaggregated Memory May 2026
Sidharth Sankhe, **Felix Zhang**, Umayrah Chonee, Sherman Lim, Jason Hu, Jialin Li, and Qizhen Zhang
23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)

Research Experience

Research Assistant; Far Data Lab, University of Toronto Sept. 2024 – Aug. 2025
Supervisor: Prof. Qizhen Zhang; Funded by: Far Data Lab

- Investigated offloading computation onto data processing unit (DPUs) to enable efficient, scalable data processing
- Implemented and parallelized the execution of *Monodepth2* in **C++**, achieving linear performance scaling with the number of threads when running on a DPU
- Built a DPU-based prefetcher *PD3* with a team of 6 which intercepts network traffic to predict and prefetch data for tiered key-value stores, eliminating the network overhead introduced when fetching entries from remote
- Designed an external service for offloading shuffle operations onto DPUs in distributed data analytics which supports both disagregated memory and storage backends

Research Assistant; University of Toronto March 2024 – Dec. 2024
Supervisor: Prof. Jack Sun

- Worked on a team of 11 to implement a pedagogical kernel *KidneyOS* in **Rust** to be used in an introductory operating systems course with **500+** students annually
- Enabled thread creation and destruction, multi-threading, pre-emptive scheduling within the thread system
- Led a team of 3 to implement POSIX-compatible syscalls and add support for running user-space executables

Research Assistant; PRISM Lab, Bloorview Research Institute June 2024 – Aug. 2024
Supervisors: Erica Floreani & Prof. Tom Chau; Funded by: FUSRP

- Curated deep-learning models from the literature on denoising electroencephalogram (EEG) data in a team of 4 and benchmarked them on the *EEGDenoiseNet* dataset
- Investigated the applicability of end-to-end transformer models to denoise EEG signals and the impact of using signals' time-frequency representation as input on model performance

Research Assistant; Biological Physics Group, University of Toronto May 2023 – May 2024
Supervisor: Prof. Anton Zilman; Funded by: Work Study Program

- Implemented a data pre-processing pipeline which processes raw cytokine data and extracts integral features
- Built a feed-forward network in **PyTorch** that predicts the cytokine dynamics of T cells in response to antigens
- On a team of 4, showed two variables are sufficient to determine cytokine concentrations because our model predicted the correct output concentration with **0.01%** error using a bottleneck layer with 2 neurons

Research Assistant; Physics Education Group, University of Toronto May 2022 – Sept. 2022
Supervisor: Prof. Carolyn Sealfon

- Created a dataset of ~**11 000** sentences from student feedback which labels whether they contain suggestions
- Compared the effectiveness of statistical and deep-learning classifiers at identifying suggestions using **scikit-learn** and **TensorFlow** respectively
- Demonstrated the efficacy of a BERT classifier at addressing this problem with it achieving an F_1 score of **0.823**

Work Experience

Teaching Assistant; University of Toronto — Sept. 2025 – Present
- Led tutorials and held office hours for *Introduction to Operating Systems*

ML Cluster Engineer; Cerebras Systems — May 2024 – Aug. 2025
- Implemented a runtime virtual memory system in **C++** with a team of 3 which pre-emptively loads data before it is accessed, allowing off-chip memory to be used for the first time with only a **10%** performance penalty
- Added support for network storage in the paging system with remote direct memory access, providing **100** GB/s read and write speeds with **10** μs latency to multiple remote servers
- Enabled the ability log and replay the network operations, decreasing the time to recreate stalls and timeouts by over **80%**, and setup unit tests to automatically catch breakages and performance regressions in the network layer
- Improved the throughput of the network layer by **6%** when transferring data by implementing best practices for remote direct memory access and reducing setup overhead
- Determined the cable and port mapping for one, two, and four rack clusters used in upcoming deploymenets

Awards & Scholarships

Ontario Graduate Scholarship ($15 000), University of Toronto — May 2025 – April 2026

Queen Elizabeth II Graduate Scholarships in Science and Technology ($15 000) (Declined), University of Toronto — May 2025 – April 2026

Fields Undergraduate Summer Research Program ($3 800), Fields Institute — June – Aug. 2024

Louis Savlov Scholarship in Sciences And Humanities ($1 000), University of Toronto — Nov. 2023 – Jan. 2025

Dean's List Scholar, University of Toronto — Jan. 2022 – June 2025

Second Malcom Wallace Scholarship ($5 000), University of Toronto — Sept. 2021 – Oct. 2024

University of Toronto Scholar ($7 500), University of Toronto — Sept. 2021

Student Leadership

Director of Internal Relations; Computer Science Student Union, University of Toronto — April 2023 – April 2024
- Organized orientation for the ~**500** undergraduate students entering the computer science stream
- Planned **20+** events in collaboration with various partners in industry (such as AMD and Google) or student organizations (such as UTMIST and WiCS)
- Hosted **5+** talks with professors in the Department of Computer Science at the University of Toronto

First-Year Academic Officer; Math Union, University of Toronto — Sept. 2021 – April 2022
- Facilitated discussions between **20** mentor-mentee pairs in the *First-Year Mentorship Program* by providing guidance to the upper-year mentors
- Organized "Coffee and Chat" events which allowed for informal discussions between students and math professors

Registered Study Group Leader; Sidney Smith Commons, University of Toronto — Sept. 2021 – April 2022
- Led study groups for *Foundations of Computer Science I* and *Enriched Introduction to the Theory of Computation*
- Headed weekly meetings for first-year students that reviewed content covered in the previous week's lecture
- Developed example problems to clarify and reinforce important concepts through group discussion

Projects

KivikDB — Sept. 2025 – Dec. 2025
- Built a key-value database in **Rust** which uses an LSM-tree in storage with in-memory Bloom filters
- Implemented the filter and leveling policies respectively introduced in the Monkey and Dostoevsky key-value stores

Image Classifier — Dec. 2022 – Jan. 2023
- Implemented a softmax classifier with stochastic gradient descent (SGD) from scratch in **C++** using only the linear algebra library **Eigen3**
- Achieved **92%** accuracy on the *MNIST* dataset of handwritten digits (within 2% of the top classifier using SGD)
- Built in the ability to save trained weights, perform batch training, and track the training error in real-time

Image Restoration with Convolutional Neural Networks — Sept. 2020 – June 2021
- Combined the models RIDNet and DeepDeblur using **PyTorch** to determine the ability of convolutional neural networks to deblur and denoise images
- Artificially generated a dataset of **5 000** noisy, blurred images using a Poisson-Gaussian noise model
- Discovered that integrating the two models offers marginal improvements over their individual performance